Combined Wazuh Agent Auto Deploy + Security Recommendation Enforcer

New Employee Operational Guide and SOP

For SOC, SIEM, Endpoint Security, and Infrastructure Teams

Document purpose: This guide explains how a new employee can safely use the combined Wazuh deployment scripts to install, configure, harden, validate, and report Wazuh Agent readiness across Windows, Linux, and macOS endpoints.

Field	Value
Script package	wazuh-combined-auto-deploy.zip
Supported OS	Windows, Linux, macOS
Primary objective	Auto-deploy Wazuh Agent and apply SOC-ready security monitoring configuration
Validation marker	WAZUH_READINESS_TEST
Main communication port	TCP 1514
Enrollment port	TCP 1515
Optional API validation port	TCP 55000

1. Executive Summary

The combined scripts automate two previously separate activities: Wazuh Agent deployment and security-readiness enforcement. Instead of asking a new employee to install the agent, edit ossec.conf, enable recovery, configure audit settings, and validate ingestion manually, the combined workflow performs these actions in a controlled sequence.

The scripts are designed for production operations, but they still require human judgement for lab approval, network firewall confirmation, FIM noise tuning, and final dashboard validation.

Automation Area	What the Script Does	Human Review Still Needed
Deployment	Installs Wazuh Agent if missing and writes SOC-ready ossec.conf	Confirm correct Wazuh Manager and endpoint naming standard
Connectivity	Checks TCP 1514, 1515, and optional 55000	Open upstream firewall/security group if blocked
Reliability	Configures service auto-start and restart-on-failure	Confirm endpoint sleep/power policy for servers
Security logging	Enables Windows audit/PowerShell logging, Linux auditd rules, macOS log paths	Review policy impact and local compliance
FIM	Adds important monitored paths and noise-control ignores	Tune after 24-48 hours based on alert volume
Validation	Generates WAZUH_READINESS_TEST event	Confirm event appears in Wazuh Dashboard

2. Who Should Use This Guide

- New SOC/SIEM engineers deploying Wazuh agents to endpoints.
- Endpoint administrators supporting Windows, Linux, and macOS monitoring.
- Infrastructure engineers validating Wazuh Manager connectivity.
- Security auditors verifying that SOC telemetry requirements are applied.

3. Package Structure

After extracting the ZIP file, the repository structure should look like this:

```
wazuh-combined-auto-deploy/
├── README.md
├── LICENSE
├── .gitignore
├── deploy-combined-menu.sh
├── docs/
│   ├── USAGE.md
│   ├── SECURITY_RECOMMENDATIONS.md
│   └── VALIDATION.md
└── scripts/
    ├── windows/
    │   └── Invoke-Wazuh-Combined-Windows.ps1
    ├── linux/
    │   └── invoke-wazuh-combined-linux.sh
    └── macos/
        └── invoke-wazuh-combined-macos.sh
```

File	Purpose
deploy-combined-menu.sh	Menu launcher for Linux/macOS users. Windows users should run the PowerShell script directly.
Invoke-Wazuh-Combined-Windows.ps1	Combined deployment, hardening, and validation script for Windows endpoints.
invoke-wazuh-combined-linux.sh	Combined deployment, hardening, auditd, and validation script for Linux endpoints.
invoke-wazuh-combined-macos.sh	Combined deployment, hardening, LaunchDaemon, and validation script for macOS endpoints.
docs/USAGE.md	Short command reference.
docs/SECURITY_RECOMMENDATIONS.md	Explains security controls integrated into the scripts.
docs/VALIDATION.md	Explains how to verify deployment success.

4. What the Combined Scripts Automate

- Ask for Wazuh Manager IP/FQDN, Agent Name, and Agent Group/Profile.
- Ask for lab or production-pilot acknowledgement before proceeding.
- Check Wazuh Manager ports: TCP 1514, TCP 1515, and optional TCP 55000.
- Install Wazuh Agent if it is not already installed.
- Write a SOC-ready ossec.conf with required log collection and telemetry settings.
- Enable Wazuh Agent reconnect tuning and local service auto-recovery.
- Check enrollment password handling and warn about plaintext password files.
- Optionally run agent-auth enrollment.
- Enable Windows audit policy and PowerShell logging.

- Check and optionally install Sysmon on Windows.
- Install and configure auditd rules on Linux.
- Configure macOS LaunchDaemon auto-start behavior.
- Generate a validation event using the WAZUH_READINESS_TEST marker.
- Restart the Wazuh Agent and save a readiness report.

5. Prerequisites Before Running the Script

Requirement	Windows	Linux	macOS
Admin/root privilege	Run PowerShell as Administrator	Run with sudo/root	Run with sudo/root
Internet access	Needed to download Wazuh Agent and optional Sysmon	Needed to install packages	Needed to download Wazuh Agent pkg
Manager reachability	TCP 1514 and 1515	TCP 1514 and 1515	TCP 1514 and 1515
Endpoint naming standard	Required for AgentName	Required for agent-name	Required for agent-name
Enrollment password	Only if manager requires it	Only if manager requires it	Only if manager requires it

- Confirm Wazuh Manager IP or FQDN before starting.
- Confirm whether your organization uses an enrollment password.
- Confirm endpoint group naming, for example: windows,soc,workstation or linux,soc,server.
- For Windows, confirm whether Sysmon should be installed automatically or manually with an approved configuration.
- For production servers, confirm that endpoint sleep/hibernate policy is disabled where appropriate.

6. Script Input Fields Explained

Input	Example	Meaning	Recommendation
Manager	10.10.10.50 or wazuh-manager.company.local	Wazuh Manager IP/FQDN used by the agent	Use FQDN if DNS is stable; use IP for isolated networks
AgentName	HR-LAPTOP-01	Name shown in Wazuh Dashboard	Follow asset naming standard
AgentGroup	windows,soc,workstation	Wazuh group/profile assigned to endpoint	Use OS + role + SOC tag
Enrollment password	********	Password used by Wazuh authd, if enabled	Enter at runtime only; do not store in scripts
InstallSysmonIfMissing	PowerShell switch	Allows Windows script to install Sysmon if absent	Use approved Sysmon config in production
NonInteractive	Script switch	Runs with fewer prompts for automation	Use only after lab testing

7. Recommended Agent Naming and Grouping

Endpoint Type	Agent Name Example	Agent Group Example
Windows workstation	HR-LAPTOP-01	windows,soc,workstation
Windows server	WIN-FILE-01	windows,soc,server
Domain controller	DC01	windows,soc,domain-controller
Linux web server	LINUX-WEB-01	linux,soc,webserver
Linux database server	LINUX-DB-01	linux,soc,database
macOS user device	MACBOOK-FIN-01	macos,soc,workstation

8. Safe Deployment Workflow for New Employees

1. Extract the ZIP package to a working folder.
2. Read README.md and this guide before running any script.
3. Confirm Wazuh Manager IP/FQDN with the SIEM team.
4. Confirm endpoint name and group/profile naming convention.
5. Run first in a lab or on one pilot endpoint.
6. Verify WAZUH_READINESS_TEST appears in Wazuh Dashboard.
7. Review FIM noise after 24-48 hours.
8. Only then proceed to wider production deployment.

9. Windows Deployment Guide

Use the Windows script when deploying to Windows workstation, Windows server, or Domain Controller endpoints. It must be executed from an elevated PowerShell session.

9.1 Interactive Windows Run

```
Set-ExecutionPolicy Bypass -Scope Process -Force
.\scripts\windows\Invoke-Wazuh-Combined-Windows.ps1
```

9.2 Windows Run with Parameters

```
.\scripts\windows\Invoke-Wazuh-Combined-Windows.ps1 `
  -Manager 10.10.10.50 `
  -AgentName HR-LAPTOP-01 `
  -AgentGroup "windows,soc,workstation" `
  -InstallSysmonIfMissing
```

9.3 Windows Non-Interactive Run

```
.\scripts\windows\Invoke-Wazuh-Combined-Windows.ps1 `
  -Manager wazuh-manager.company.local `
  -AgentName HR-LAPTOP-01 `
  -AgentGroup "windows,soc,workstation" `
  -NonInteractive `
  -InstallSysmonIfMissing
```

9.4 What the Windows Script Configures

- Installs Wazuh Agent MSI if the agent is missing.
- Writes C:\Program Files (x86)\ossec-agent\ossec.conf.
- Uses TCP 1514 for reliable manager communication.
- Enables Security, System, Application, PowerShell, Sysmon, Defender, RDP, Task Scheduler, WMI, WinRM, Firewall, DNS, and optional DC logs.
- Enables Windows audit policy for logon, account management, process creation, policy change, and privilege use.
- Enables PowerShell Script Block Logging and Module Logging.
- Configures WazuhSvc automatic startup and restart-on-failure.
- Generates Windows Application event ID 9001 using source WazuhReadiness.

10. Linux Deployment Guide

Use the Linux script for Ubuntu, Debian, RHEL, CentOS, Rocky Linux, and AlmaLinux endpoints. It installs prerequisites, Wazuh Agent, auditd, SOC audit rules, and systemd recovery settings.

10.1 Interactive Linux Run

```
chmod +x deploy-combined-menu.sh
./deploy-combined-menu.sh
```

10.2 Direct Linux Run

```
sudo bash ./scripts/linux/invoke-wazuh-combined-linux.sh
```

10.3 Linux Run with Parameters

```
sudo bash ./scripts/linux/invoke-wazuh-combined-linux.sh \
  --manager 10.10.10.50 \
  --agent-name linux-web-01 \
  --agent-group linux,soc,webserver
```

10.4 Linux Non-Interactive Run

```
sudo bash ./scripts/linux/invoke-wazuh-combined-linux.sh \
  --manager wazuh-manager.company.local \
  --agent-name linux-web-01 \
  --agent-group linux,soc,webserver \
  --non-interactive
```

10.5 What the Linux Script Configures

- Installs Wazuh Agent from official Wazuh repositories if missing.
- Writes /var/ossec/etc/ossec.conf.
- Collects auth, secure, syslog, messages, auditd, journald, web server, database, Docker, firewall, and Wazuh internal logs.
- Installs and enables auditd.
- Writes SOC audit rules to /etc/audit/rules.d/soc.rules.
- Configures systemd Restart=always for wazuh-agent.
- Generates a syslog event with WAZUH_READINESS_TEST.

11. macOS Deployment Guide

Use the macOS script for Mac endpoints. It installs the Wazuh pkg if missing, writes macOS SOC log collection settings, loads the LaunchDaemon, and generates a syslog validation event.

11.1 Interactive macOS Run

```
chmod +x deploy-combined-menu.sh
./deploy-combined-menu.sh
```

11.2 Direct macOS Run

```
sudo bash ./scripts/macos/invoke-wazuh-combined-macos.sh
```

11.3 macOS Run with Parameters

```
sudo bash ./scripts/macos/invoke-wazuh-combined-macos.sh \
  --manager 10.10.10.50 \
```

```
--agent-name macbook-finance-01 \
--agent-group macos,soc
```

11.4 What the macOS Script Configures

- Installs Wazuh Agent pkg if missing.
- Writes /Library/Ossec/etc/ossec.conf.
- Collects system.log, install.log, wifi.log, audit logs, application logs, diagnostic logs, and Wazuh internal logs.
- Monitors macOS persistence paths such as LaunchDaemons, LaunchAgents, PrivilegedHelperTools, SSH, and user LaunchAgents.
- Loads the Wazuh LaunchDaemon.
- Generates a syslog event with WAZUH_READINESS_TEST.

12. Security Recommendation Automation Mapping

Recommendation	How the Combined Script Handles It	Expected Employee Action
Test scripts in a lab	Creates a lab acknowledgement marker after confirmation	Run first in a lab or pilot endpoint
Confirm Wazuh Manager ports are open	Tests TCP 1514, 1515, and optional 55000	Escalate to network/firewall team if blocked
Confirm enrollment password handling policy	Searches common paths for plaintext password files	Use runtime prompt; do not save password in files
Review FIM paths for noise	Adds recommended ignore paths and writes warning	Review Wazuh alerts after 24-48 hours
Confirm Sysmon is installed on Windows	Checks service and optionally installs Sysmon	Use approved Sysmon config for production
Confirm auditd is installed on Linux	Installs/enables auditd and applies SOC rules	Validate /var/log/audit/audit.log ingestion
Validate event ingestion	Generates WAZUH_READINESS_TEST	Search marker in Wazuh Dashboard

13. SOC-Ready ossec.conf Overview

The scripts overwrite ossec.conf after taking a timestamped backup. The new configuration includes connectivity, buffering, log collection, FIM, rootcheck, SCA, syscollector, and active response baseline settings.

Section	Purpose
client	Defines Wazuh Manager address, TCP 1514, notify_time, reconnect, and profile.
client_buffer	Buffers events during temporary network/manager interruption.
localfile	Collects OS, security, application, audit, and service logs.
syscheck	File Integrity Monitoring for critical paths and persistence locations.
rootcheck	Checks for rootkits, suspicious files, ports, and system indicators.
sca	Runs Security Configuration Assessment checks.
syscollector	Collects inventory: OS, packages, ports, processes, network, hardware.

14. Validation After Deployment

14.1 Universal Dashboard Validation

After any deployment, open Wazuh Dashboard and search for:

```
WAZUH_READINESS_TEST
```

For Windows, also search for:

```
WazuhReadiness
event.id: 9001
```

14.2 Agent Status Validation

OS	Command
Windows	Get-Service WazuhSvc
Linux	sudo systemctl status wazuh-agent
macOS	sudo /Library/Ossec/bin/wazuh-control status

14.3 Network Validation

```
# Windows
Test-NetConnection WAZUH_MANAGER -Port 1514
Test-NetConnection WAZUH_MANAGER -Port 1515

# Linux/macOS
nc -vz WAZUH_MANAGER 1514
nc -vz WAZUH_MANAGER 1515
```

14.4 Agent Log Validation

```
# Windows
Get-Content "C:\Program Files (x86)\ossec-agent\ossec.log" -Tail 50

# Linux
sudo tail -n 50 /var/ossec/logs/ossec.log

# macOS
sudo tail -n 50 /Library/Ossec/logs/ossec.log
```

15. Readiness Report Locations

OS	Report Location
Windows	C:\ProgramData\WazuhCombinedDeploy\
Linux	/var/log/wazuh-combined-deploy/
macOS	/Library/Logs/WazuhCombinedDeploy/

The report records all OK, FIXED, WARN, and FAIL results. Attach this report to the deployment ticket or change request.

16. Troubleshooting Guide

Issue	Likely Cause	Fix

Agent disconnected	Manager unreachable, service stopped, bad key, firewall block	Check service, TCP 1514, ossec.log, and re-enroll if needed
TCP 1514 failed	Firewall/security group/network path blocked	Open TCP 1514 agent-to-manager
TCP 1515 failed	Enrollment service/port blocked	Open TCP 1515 or skip enrollment if already registered
Sysmon not found	Sysmon not installed	Install with approved config or use -InstallSysmonIfMissing
auditd not logging	auditd not running or rules not loaded	Run auditctl -l and check /etc/audit/rules.d/soc.rules
No WAZUH_READINESS_TEST in dashboard	Agent not connected, log channel missing, indexing delay	Check agent logs and wait a few minutes; verify Wazuh Dashboard search time range
Too many FIM alerts	Monitored path too noisy	Tune syscheck paths and ignores based on endpoint role

17. Re-Enrollment Procedure

Use re-enrollment when the agent key is invalid, agent identity is duplicated, or the manager shows the endpoint as disconnected despite local service health.

```
# Windows
Stop-Service WazuhSvc
& "C:\Program Files (x86)\ossec-agent\agent-auth.exe" -m WAZUH_MANAGER -A UNIQUE_AGENT_NAME
Restart-Service WazuhSvc

# Linux
sudo systemctl stop wazuh-agent
sudo /var/ossec/bin/agent-auth -m WAZUH_MANAGER -A UNIQUE_AGENT_NAME
sudo systemctl restart wazuh-agent

# macOS
sudo /Library/Ossec/bin/wazuh-control stop
sudo /Library/Ossec/bin/agent-auth -m WAZUH_MANAGER -A UNIQUE_AGENT_NAME
sudo /Library/Ossec/bin/wazuh-control start
```

18. New Employee SOP Checklist

- **[ ]** Confirm the target OS and hostname.
- **[ ]** Confirm Wazuh Manager IP/FQDN.
- **[ ]** Confirm TCP 1514 and 1515 are allowed from endpoint to manager.
- **[ ]** Confirm agent group/profile naming standard.
- **[ ]** Confirm whether enrollment password is required.
- **[ ]** Run the combined script as Administrator/root.
- **[ ]** Save the readiness report to the deployment ticket.
- **[ ]** Verify agent appears Active in Wazuh Dashboard.
- **[ ]** Search for WAZUH_READINESS_TEST.
- **[ ]** For Windows, confirm Sysmon channel events if Sysmon is required.
- **[ ]** For Linux, confirm auditd events in /var/log/audit/audit.log.
- **[ ]** Review FIM alerts after 24-48 hours.
- **[ ]** Document any WARN/FAIL items and escalate if needed.

19. Operational Warnings and Best Practices

- Do not store enrollment passwords in plain text files, GitHub repositories, tickets, or screenshots.
- Do not mass deploy the script before a lab or pilot test.
- Do not enable extremely broad realtime FIM on high-churn directories without testing.
- Do not assume local firewall rules fix upstream firewalls; network devices may still block traffic.
- Do not ignore WARN or FAIL lines in the readiness report.
- Use approved Sysmon configuration in production rather than relying only on the minimal fallback config.
- Coordinate with server owners before deploying to critical production systems.

20. Frequently Asked Questions

Can the script guarantee the agent never disconnects?

No. It makes the agent auto-starting and auto-recovering when endpoint and network are available. It cannot prevent disconnects caused by power-off, sleep, network outage, firewall block, or Wazuh Manager outage.

Should I use Manager IP or FQDN?

Use FQDN when DNS is reliable and easier to maintain. Use IP for isolated environments or where DNS is not stable.

What if TCP 1514 fails?

The endpoint cannot reliably send logs to the manager. Open the firewall or fix routing before production deployment.

What if TCP 1515 fails?

Enrollment may fail. If the agent is already enrolled, 1515 may not be required continuously, but it is required for new enrollment.

What if the dashboard does not show WAZUH_READINESS_TEST?

Check time range, agent status, local ossec.log, connectivity, and whether the correct log source is being collected.

Should every endpoint install Sysmon?

For Windows SOC visibility, Sysmon is strongly recommended, but production rollout should use an approved and tested Sysmon config.

Will the script remove old ossec.conf?

No. It creates a timestamped backup before writing the new SOC-ready configuration.

21. Quick Command Reference

```
# Windows interactive
Set-ExecutionPolicy Bypass -Scope Process -Force
.\scripts\windows\Invoke-Wazuh-Combined-Windows.ps1

# Windows parameterized
.\scripts\windows\Invoke-Wazuh-Combined-Windows.ps1 `
  -Manager 10.10.10.50 `
  -AgentName HR-LAPTOP-01 `
  -AgentGroup "windows,soc,workstation" `
```

```
  -InstallSysmonIfMissing

# Linux interactive
sudo bash ./scripts/linux/invoke-wazuh-combined-linux.sh

# Linux parameterized
sudo bash ./scripts/linux/invoke-wazuh-combined-linux.sh \
  --manager 10.10.10.50 \
  --agent-name linux-web-01 \
  --agent-group linux,soc,webserver

# macOS interactive
sudo bash ./scripts/macos/invoke-wazuh-combined-macos.sh

# macOS parameterized
sudo bash ./scripts/macos/invoke-wazuh-combined-macos.sh \
  --manager 10.10.10.50 \
  --agent-name macbook-finance-01 \
  --agent-group macos,soc
```